NATIONAL UNIVERSITY OF MODERN LANGUAGES

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SUBMITTED TO:

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Subject: ……………………………………….. OOPs

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Section: …………………………………………. BSSE-II-A-afternoon

Assignment # 02

CLO 2: Identify the objects & their relationships to build object oriented solution.

Question #1: Encapsulation is one of the four fundamental concepts of Object-oriented programming. Identify the importance of encapsulation in object oriented programming and also implement an example (scenario/code) to build an object oriented solution using encapsulation concept.

Importance of encapsulation:

- When maintaining code, encapsulation helps us protect many parts of the program from extensive modification if the design decision is changed, once we hide our implementation from its client and do it properly, changing that implementation will not affect the client's code, which can help us avoid the risk of breaking things that are not related to the change we are making.
- One more point about maintainability, a well-encapsulated code is much easier to read, understand with minimum waste of time. Imagine if you are supposed to add a feature on top of code that you are not familiar with, and you want to reuse that code, if the code is not encapsulated well, you may end up reading most of the functions/classes before you use them, and be sure to manage their state properly, so this can cause developers to waste a lot of time reading the current code, instead of writing new code.
- As mentioned earlier, encapsulation also protects the integrity of our code, and this is done by not allowing the client to modify the internal data and put the component in an invalid state, which makes the code more safe and reliable, in addition, if the fail-fast mechanism is used properly, developers can detect failures early and get enough information about the error and how to avoid it.
- It helps us reduce the complexity of our design, by reducing the tight coupling between different components in our code, and it can help reduce spaghetti code and also helps developers to be more productive.
- Encapsulation also makes it easier to unit-test our code, because it helps us encapsulate and manage all the properties and methods related to each other in one place, which makes it easier for us to test them.

Getters and Setters

Getters and Setters in java are two methods used for fetching and updating the value of a variable**.** Getter methods are concerned with fetching the updated value of a variable, while a setter method is used to set or update an existing variable's value

Example

```
class Student{

private String name;

public String getName()

    {

    return name;

    }

public void setName(String name)

    {

    this.name=name;

    }



    }

public class Test{

    public static void main(String[] args)

    {
```

```
Student s=new Student();

s.setName("hello world");

System.out.println(s.getName());

}

}
```

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS E:\java> cd "e:\java\" ; if ($?) { javac tut18.java } ; if ($?) { java tut18 }
hello world
PS E:\java>
```